

Agniveer Vayu Other than Science (Group Y) - 14 Oct 2023 - Shift-1 - Memory Based Paper

English

Question 1

Select the most appropriate synonym of the given word.

Eternal

Options:

- A. Perpetual
- B. Repetition
- C. Prolonged
- D. Look

Answer: A

Solution:

The correct answer is '**Perpetual**'.

Key Points

- The word '**Eternal**' means 'lasting forever or for a very long time (हमेशा के लिए या बहुत लंबे समय तक चलने वाला)'.
- The **synonyms** of the word '**Eternal**' are "abiding, ageless, perpetual, continuing, dateless, enduring, everlasting, immortal, imperishable, lasting, ongoing, perennial, perpetual, timeless, undying".
- **From the synonym** of the given word, we can say that the word '**Perpetual**' is the **most similar in meaning**.
- The word '**Perpetual**' means 'continuing forever in the same way (उसी तरह से हमेशा के लिए जारी)'.

Hence, the correct answer is **option 1**. Let's see the **meaning** of other given options:-

Words	Meaning
Repetition	the act of doing or saying something again (फिर से कुछ करने या कहने की क्रिया)

Prolonged	extend the duration of.
Look	to direct your eyes in order to see (देखने के लिए अपनी आंखों को निर्देशित करने के लिए)

Additional Information

- The **antonym** of the word '**Eternal**' is "transient".
- **Example of 'Eternal' in a sentence:**
 - *Will you two never stop your **eternal** arguing!*

Question 2

What is the antonym of the given word?

Gentle

Options:

- A. Mild
- B. Tender
- C. Harsh
- D. Merciful

Answer: C

Solution:

The correct answer is "**Harsh**".

Key Points

- '**Gentle**' denotes someone or something that is mild and kind, and does not cause harm or discomfort. (सौम्य)
 - **Example**-The deer in the forest were surprisingly **gentle**.
- '**Harsh**' refers to something that is unkind, cruel, or more severe than is necessary. (कठोर)
 - **Example**-Her teacher's **harsh** words dampened her enthusiasm for learning.
- Hence, the option that is Opposite in meaning to the given word is 'Harsh'.

Therefore, the correct answer is '**Option 3**'.

Additional Information

- Let us have a look at the meaning of the remaining options.
 - 'Mild' implies a lack of harshness or severity.(सौम्य)
 - 'Tender' generally refers to something soft or delicate. In another context, it could also mean showing gentleness, kindness, and affection.(कोमल)
 - 'Merciful' denotes showing or exercising mercy. It refers to someone who forgives easily or is kind and forgiving.(दयालु)
-

Question 3

In the following question, out of the given four alternatives, select the alternative which best expresses the meaning of the Idiom/Phrase.

In black and white

Options:

- A. In writing
- B. In trouble
- C. In suspense
- D. In brief

Answer: A

Solution:

The correct answer is 'Option 1'.

Key Points

- "In black and white", the correct idiom meaning is "In writing" (कागज़ी रूप से).
- It refers to having something documented or written down.

Therefore, the correct answer is: 'In writing'

Additional Information

- In trouble(मुसीबत में) indicates a problematic situation, which doesn't match with the given idiom.
- In suspense(सस्पेंस में) refers to a state of uncertainty or excitement, again, not connected to the idiom.

- **In brief**(संक्षेप में) means something that is short and concise, not matching with the idiom.
-

Question 4

Find the word which is correctly spelt from the given options.

Options:

- A. Exemplary
- B. Indifferance
- C. Heirarchy
- D. Feudel

Answer: A

Solution:

The correct answer is, '**Exemplary**'.



Key Points

- '**Exemplary**' is the only correct spelling in the given options that means something/someone deserving of honour.
- Hence, the correct answer is '**option 1**'

Therefore, the correctly spelt word is - '**Exemplary**'.



Additional Information

- The correct spelling of the given words with their meaning.
 - **Indifference**- lack of enthusiasm.
 - **Hierarchy**- ranking, grading.
 - **Feudal**-relating to the social system of western Europe in the Middle Ages
-

Question 5

Which of the following words is correctly spelled?

Options:

A. Embraice

B. Embrase

C. Imbrace

D. Embrace

Answer: D

Solution:

The correct answer is “**Embrace**”



Key Points

- "Embrace" is spelled E-M-B-R-A-C-E.
- The word "Embrace" means to clasp in the arms, cherish, love.

Hence, the correct answer is '**Option 4**'.

Question 6

What is the verb form of the given noun?

Expansion

Options:

A. Expandable

B. Expanding

C. Expand

D. Expansionist

Answer: C

Solution:

The correct answer is "**Expand**".

Key Points

- 'Expansion' is a noun denoting the action, process, or result of expanding.
- The **verb** form, which signifies the act of expanding something or becoming more extensive, is 'expand.'
- Hence, the correct answer is '**Option 3**'.

Additional Information

- 'Expandable' is an adjective resulting from the verb 'expand,' denoting the characteristic of being able to be made larger or wider. This word describes the possibility or ability of something to expand, not the action itself.
 - 'Expanding' is the present participle of the verb 'expand.' It represents an ongoing action of becoming larger or more extensive. Though it originates from the same verb, this is not the base verb form.
 - 'Expansionist' signifies an advocate of geographical or economic expansion. Even though it comes from the same root word, it is not the action form of the word 'expansion.'
-

Question 7

Convert the given sentence into Indirect speech.

Sunil said, "Where is Sunita?"

Options:

- A. Sunil asked where Sunita had been
- B. Sunil asked where Sunita is
- C. Sunil asked where Sunita was.
- D. Sunil said that where is Sunita.

Answer: C

Solution:

The correct answer is "Sunil asked where Sunita was."

Key Points

- When turning a direct speech into an indirect one, we have to consider tense changes. The original question Sunil poses is in the present tense. However, because we're expressing this as indirect speech, it must be moved back a tense to become past tense - hence the "**is**" from the original sentence changes to "**was**".

- The punctuation used in the sentence is also crucial. In the direct speech, Sunil's statement is a question, indicated by the question mark. Once we change it to indirect speech, it becomes a statement about what Sunil asked. Consequently, the question mark should be replaced with a period in the indirect form.
- The form of the sentence changes as well in indirect speech. In direct speech, the question "where is Sunita?" is asked straightforwardly. However, in the indirect form, the sentence is more complex - it has to indicate that Sunil is the one who asked the question and what exactly the question was, all in a single statement. Therefore, the correct form should be "**Sunil asked where Sunita was.**"
- None of the other options align perfectly with these rules and considerations. Option A misgenders the tenses by using "had been" instead of "was," Option B retains the present tense "is," which is incorrect, and Option D fails to structure the indirect statement correctly, for it retains the "where is" order from the direct speech which is not appropriate for indirect speech.

Hence, the correct answer is '**Option 3**'.

Question 8

Transform the following sentence from active voice to passive voice:

Someone is following us

Options:

- A. We are being followed by someone
- B. Someone has been followed by us
- C. We were followed by someone.
- D. Someone are followed by us.

Answer: A

Solution:

The correct answer is "**We are being followed by someone.**"



Key Points

- The original sentence "Someone is following us" is in the active voice. The active voice in a sentence is the one where the subject performs the action. In this case, "Someone" performs the action of "following."
- In passive voice, the object of the active verb becomes the subject of the passive verb. Therefore, in transforming the sentence into passive, "**us**" becomes the subject, making "**we**" the proper word to start the sentence.
- The verb tense should remain the same when converting from active to passive voice. The active sentence uses present continuous tense ("**is following**"), so the passive form also needs to use present continuous

tense. The passive form of present continuous is "**are being followed**" for "we", maintaining the same tense.

- The word "someone," which was the subject of the active sentence, now turns into the agent in the passive voice that follows the action "are being followed," retained by the use of the preposition "**by.**"
- Hence, Option 1, "We are being followed by someone," correctly transforms the active sentence into the passive voice without changing the meaning of the sentence.

Hence, the correct answer is '**Option 1**'.

Question 9

Fill in the blank with correct option:

It _____ if you had not caught it.

Options:

- A. will break
- B. was breaking
- C. would have broken
- D. would break

Answer: C

Solution:

The correct answer is **would have broken**.



Key Points

- The given sentence is a third conditional type of question. In the third conditional sentence, the structure of the sentence is-
 - **If + Subject + had + V₃ + Object, Subject + Would + Have + V₃.**
- In the If-clause the structure used is "had not caught" this gives us a hint that this is the third type of conditional sentence.
 - Example-If you **had told** me about the meeting, I **would have come**.

So the correct sentence is- It would have broken if you had not caught it.

Question 10

Choose the correct conjunction to fill in the blank:

Do you want tea _____ coffee?

Options:

- A. or
- B. and
- C. yet
- D. but

Answer: A

Solution:

The correct answer is "**Or**".



Key Points

- The conjunction "or" is typically used to present alternatives or choices in a sentence.
- In this case, a choice between two items (tea and coffee) is being offered.
- The conjunction "or" is used to express that only one of the two options can be chosen.
- Therefore, the use of "or" is the most appropriate for this sentence.
- The correct sentence is- **Do you want tea or coffee?**

Hence, the correct answer is '**Option 1**'.

Question 11

Choose the correct verb for the blank.

Look there! the robbers _____ away.

Options:

- A. run

B. have been running

C. are running

D. ran

Answer: C

Solution:

The correct option is: **are running**.



Key Points

- The **present continuous tense** (are running) is appropriate here because the speaker is discussing an action that is happening right now, in the real time of the speaker and the listener.
- In English, the present continuous tense is formed by using the verb '**to be**' (in this case, 'are' for 'the robbers') followed by a **present participle**—an 'ing' verb (in this case, 'running').
- Since the sentence is asking someone to look at action occurring right now, this is the appropriate tense to use.

Therefore, the correct answer is: *Look there! The robbers **are running** away.*



Additional Information

- **"run"** - This is the base form of the verb, which is typically not used to describe an action happening right now. It might be used in a general sense as in "Robbers run away after a theft."
- **"have been running"** - This is the present perfect continuous tense, used to describe an ongoing action that started in the past and is still continuing or has just finished. In this case, "The robbers have been running away" would imply an ongoing or recently completed action, but it wouldn't be the proper tense to describe a new action currently being observed.
- **"ran"** - This is the past simple tense and is used to describe completed actions in the past. The sentence "The robbers ran away" indicates that the action is already finished, which doesn't match the command "Look there!" indicating a current ongoing event.

Question 12

Improve the underlined part. If no improvement is required, mark your answer as 'No improvement'.

When I get home, my children would be playing.

Options:

A. are playing

- B. were playing
- C. will have been playing
- D. No improvement

Answer: C

Solution:

The correct answer is **“will have been playing”**.

Key Points

- The original sentence, **"When I get home, my children would be playing"** uses the **incorrect tense**. The phrase "would be playing" is the conditional continuous tense. It's used to describe actions that would happen under certain conditions. However, in the context of the sentence, the action is not conditional but rather a future prediction.
- Option 1 "are playing", makes the sentence present continuous - "When I get home, my children are playing". This is incorrect because it inaccurately suggests that the children are currently playing at the time of speaking, rather than predicting what they will be doing in the future when the speaker arrives home.
- Option 2 "were playing", makes the sentence past continuous - "When I get home, my children were playing". This is incorrect as it suggests the children were playing at the same time in the past when the speaker was getting home; it doesn't align with the future event of the speaker getting home.
- Option 3 **"will have been playing"**, makes the sentence future perfect continuous - "When I get home, my children will have been playing". This tense is used to describe an ongoing action that will be in progress up to some point in the future. Here, it suggests that the children will start playing before the speaker's arrival and continue to do so when the speaker arrives, which correctly matches the future context given in the sentence.

Hence, the correct answer is **‘Option 3’**.

Question 13

In the given sentence, identify the segment which contains the grammatical error

Many a man have realized that real happiness lies in making sacrifices.

Options:

- A. Many a man

- B. have realized that
- C. real happiness lies
- D. in making sacrifices.

Answer: B

Solution:

The correct answer is '**Have realized that**'.



Key Points

- The given sentence is grammatically incorrect.
- Here, '**Has realized that**' should be used instead of '**Have realized that**'.
- With "**many a**" **singular verb** is used.
- Hence we should use '**has**' and not '**have**' when **asingular verb** is required in a sentence.
- *Example:*
 - *Many a man go to park every morning in my locality. (Incorrect)*
 - *Many a man goes to park every morning in my locality. (Correct)*
- Therefore, as per the points mentioned above, we find that the correct answer is **Option 2**.

Correct Sentence: *Many a man has realized that real happiness lies in making sacrifices.*



Additional Information

- **Each, every, everyone, somebody, someone, many a, more than one** etc. are **singular**.
- Hence, they take singular verbs, singular nouns, and singular pronouns.
- *Example:*
 - *Many a student have not done their class work. (Incorrect)*
 - *Many a student has not done his class work. (Correct)*

Question 14

Select the option that can be used as a one-word substitute for the given group of words.

An incidence causing damage or suffering

Options:

A. Apostrophe

B. Philanthropy

C. Catastrophe

D. Commencement

Answer: C

Solution:

The correct answer is '**Catastrophe**'.



Key Points

- The group of words "An incidence causing damage or suffering" refers to a sudden event causing significant harm or suffering. (आपदा).
 - Example: The oil spill was a catastrophe for the local wildlife.
- 'Catastrophe' refers to a large-scale disaster, misfortune, or failure. (आपदा).
 - Example: Losing their largest client was a catastrophe for the company.

Therefore, the correct answer is '**Option 3**'.



Additional Information

- 'Apostrophe' (अपोस्ट्रोफी) is a punctuation mark used to indicate either possession or the omission of letters or numbers.
 - 'Philanthropy' (परोपकारिता) refers to the desire to promote the welfare of others, typically through donation and charitable activities.
 - 'Commencement' (आरंभ) signifies the beginning or start of something.
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Question 15

Sentences of a paragraph are given below in jumbled order. Arrange the sentences in the correct order to form a meaningful and coherent paragraph.

P. The need of the hour is to protect and preserve the environment and the ecological balance.

Q. This menace can be fought against only by creating awareness among the people.

R. Rapid deforestation has been the main cause of environmental degradation.

S. Degradation of the environment continues unabated.

Options:

A. SPQR

B. PQRS

C. QRPS

D. QPSR

Answer: A

Solution:

The correct answer is 'Option 1'.



Key Points

- The sentence 'S' sets the tone of the paragraph by mentioning the continuous degradation of the environment.
- The sentence 'P' further talks about the urgent need in response to the problem mentioned in 'S', thus creating a logical connection.
- The sentence 'Q' is about the cause, i.e., the rapid deforestation which leads to environmental degradation. Following the statement of the problem and the need, it's natural to discuss the cause.
- 'R', then provides the possible solution to the stated problem - creating awareness. This fits logically after the problem and its cause have been mentioned.

Therefore, the correct sequence is **SPQR**.



Additional Information

- The option '2' PQRS does not logically follow as it starts with a statement about the need of the hour, but doesn't describe what the problem is until we reach 'S'.
- Option '3' QRPS seems to start from the middle where it talks about the cause and remedy, yet we do not see the initial mention of a problem until 'P' and 'S'.
- '4' QPSR can't be the correct answer as it starts with a solution to fight against deforestation but the problem is addressed later in 'P' and 'S' which breaks the flow of the paragraph.

Question 16

Read the following information carefully and answer the given questions.

Octopuses are famous for their alien-like abilities, from regrowing damaged arms to changing their skin colour and texture. They use this colour-shifting power to camouflage and, interestingly, as a strange visual language to talk to other octopuses.

A little known fact is they actually belong to a category of animals (phylum) called Mollusca, which is largely made up of snails. Yep, octopuses are like souped-up snails who lost their shells and grew a rather large brain. The greatest thing about them is their intelligence, which evolved completely independently from our own.

Let's consider their nervous system. Like us, octopus have large brains compared to their body size - easily the biggest of all invertebrates (animals without a backbone) and of comparable size to many vertebrates, such as frogs.

It is, however, hard to compare brain size between marine animals and land animals, because the laws of physics differ in water and air. Animals are weightless in water but on land body shape and size is limited by gravity. An octopus brain is made up of about 500 million brain cells (neurons). This is seven times more than a mouse and about the same as a marmoset monkey. Humans, on the other hand, have 86 billion brain cells.

Testing octopus intelligence can be a problem, because the animals frequently outsmart scientists. For example, scientists can struggle to get an octopus to solve a maze, because they often climb out and crawl over the top to reach their food reward. And that's assuming they haven't already escaped from their aquarium home and are crawling around the lab.

Unlike us though, octopuses don't live for very long. The giant Pacific octopus might live up to five years, but most live for just a year and some as little as six months. They hatch from eggs fully formed and ready to go. They never see their parents and have to learn everything on their own.

Why testing octopus intelligence can be a problem?

Options:

- A. Because they can outsmart scientists.
- B. Because of their body shape and size.
- C. As they have so many legs.
- D. Because they have a very large brain.

Answer: A

Solution:

The correct answer is **option (1) Because they can even outsmart scientists.**



Key Points

- Refers to line 1 of paragraph 5, "**Testing octopus intelligence can be a problem**, because the animals frequently **outsmart scientists**."
 - Thus, **option 1 is the correct** option as per passage.

Thus, testing octopus intelligence can be a problem as given in chosen option (1) because they can outsmart scientists.

Question 17

Read the following information carefully and answer the given questions.

Octopuses are famous for their alien-like abilities, from regrowing damaged arms to changing their skin colour and texture. They use this colour-shifting power to camouflage and, interestingly, as a strange visual language to talk to other octopuses.

A little known fact is they actually belong to a category of animals (phylum) called Mollusca, which is largely made up of snails. Yep, octopuses are like souped-up snails who lost their shells and grew a rather large brain. The greatest thing about them is their intelligence, which evolved completely independently from our own.

Let's consider their nervous system. Like us, octopus have large brains compared to their body size - easily the biggest of all invertebrates (animals without a backbone) and of comparable size to many vertebrates, such as frogs.

It is, however, hard to compare brain size between marine animals and land animals, because the laws of physics differ in water and air. Animals are weightless in water but on land body shape and size is limited by gravity. An octopus brain is made up of about 500 million brain cells (neurons). This is seven times more than a mouse and about the same as a marmoset monkey. Humans, on the other hand, have 86 billion brain cells.

Testing octopus intelligence can be a problem, because the animals frequently outsmart scientists. For example, scientists can struggle to get an octopus to solve a maze, because they often climb out and crawl over the top to reach their food reward. And that's assuming they haven't already escaped from their aquarium home and are crawling around the lab.

Unlike us though, octopuses don't live for very long. The giant Pacific octopus might live up to five years, but most live for just a year and some as little as six months. They hatch from eggs fully formed and ready to go. They never see their parents and have to learn everything on their own.

A marmoset monkey has _____ brain cells.

Options:

- A. 86 billion
- B. less than 86 million
- C. 78 million
- D. about 500 million

Answer: D

Solution:

The correct answer is option (4) **about 500 million**

Key Points

- Refers to line 3 of paragraph 4, "An **octopus** brain is made up of **about 500 million brain cells** (neurons)."
- Refers to line 4 of paragraph 4, "**This** is seven times more than a mouse and **about the same as a marmoset monkey**."
 - Hence it is clear from the passage that the **marmoset** monkey has **about same number of** brain cells as an **octopus**, i.e. about 500 million.

Thus, the correct answer is option (4) about 500 million.

Question 18

Read the following information carefully and answer the given questions.

Octopuses are famous for their alien-like abilities, from regrowing damaged arms to changing their skin colour and texture. They use this colour-shifting power to camouflage and, interestingly, as a strange visual language to talk to other octopuses.

A little known fact is they actually belong to a category of animals (phylum) called Mollusca, which is largely made up of snails. Yep, octopuses are like souped-up snails who lost their shells and grew a rather large brain. The greatest thing about them is their intelligence, which evolved completely independently from our own.

Let's consider their nervous system. Like us, octopuses have large brains compared to their body size - easily the biggest of all invertebrates (animals without a backbone) and of comparable size to many vertebrates, such as frogs.

It is, however, hard to compare brain size between marine animals and land animals, because the laws of physics differ in water and air. Animals are weightless in water but on land body shape and size is limited by gravity. An octopus brain is made up of about 500 million brain cells (neurons). This is seven times more than a mouse and about the same as a marmoset monkey. Humans, on the other hand, have 86 billion brain cells.

Testing octopus intelligence can be a problem, because the animals frequently outsmart scientists. For example, scientists can struggle to get an octopus to solve a maze, because they often climb out and crawl over the top to reach their food reward. And that's assuming they haven't already escaped from their aquarium home and are crawling around the lab.

Unlike us though, octopuses don't live for very long. The giant Pacific octopus might live up to five years, but most live for just a year and some as little as six months. They hatch from eggs fully formed and ready to go. They never see their parents and have to learn everything on their own.

Identify the correct statement from the following.

I. Octopuses belong to a category of animals called Mollusca.

II. It is hard to compare brain size between marine animals and land animals.

III. Vertebrates are animals without a backbone.

Options:

- A. I and II
- B. I, II and III
- C. Only III
- D. II and III

Answer: A

Solution:

The correct answer is **option (1) I and II.**



Key Points

- Refer to the following lines from the passage: -
 - Line 1 of paragraph 2, "*A little known fact is **they actually belong to a category of animals (phylum) called Mollusca.***"
 - Thus, **statement I is correct** as per the passage.
 - Line 1 of paragraph 4, "*It is, however, **hard to compare brain size between marine animals and land animals**, because the laws of physics differ in water and air.*"
 - Thus, **statement II is correct** as per the passage.
 - Line 2 of paragraph 3, "*easily the biggest of all **invertebrates (animals without a backbone)** and of comparable size to many vertebrates.*"
 - Thus, **statement III is incorrect** as per the passage.

Therefore, the correct statements are option (1) **I and II.**

Question 19

Read the following information carefully and answer the given questions.

Octopuses are famous for their alien-like abilities, from regrowing damaged arms to changing their skin colour and texture. They use this colour-shifting power to camouflage and, interestingly, as a strange visual language to talk to other octopuses.

A little known fact is they actually belong to a category of animals (phylum) called Mollusca, which is largely made up of snails. Yep, octopuses are like souped-up snails who lost their shells and grew a rather large brain. The greatest thing about them is their intelligence, which evolved completely independently from our own.

Let's consider their nervous system. Like us, octopus have large brains compared to their body size - easily the biggest of all invertebrates (animals without a backbone) and of comparable size to many vertebrates, such as

frogs.

It is, however, hard to compare brain size between marine animals and land animals, because the laws of physics differ in water and air. Animals are weightless in water but on land body shape and size is limited by gravity. An octopus brain is made up of about 500 million brain cells (neurons). This is seven times more than a mouse and about the same as a marmoset monkey. Humans, on the other hand, have 86 billion brain cells.

Testing octopus intelligence can be a problem, because the animals frequently outsmart scientists. For example, scientists can struggle to get an octopus to solve a maze, because they often climb out and crawl over the top to reach their food reward. And that's assuming they haven't already escaped from their aquarium home and are crawling around the lab.

Unlike us though, octopuses don't live for very long. The giant Pacific octopus might live up to five years, but most live for just a year and some as little as six months. They hatch from eggs fully formed and ready to go. They never see their parents and have to learn everything on their own.

What is the synonym of the word-'strange' as used in the passage?

Options:

A. Conventional

B. Vulnerable

C. Peculiar

D. Deify

Answer: C

Solution:

The correct answer is **option (3) Peculiar**.



Key Points

- In the given passage, "*as a **strange** visual language to talk to other octopuses*" strange means **different from what is usual or unexpected, which makes you feel slightly nervous or afraid** (असामान्य, अनूठा या अप्रत्याशित, जो आपको थोड़ा घबराहट या डर महसूस कराता है)
- In the given options: -
 - **Conventional** means **behavior or opinion that are ordinary and normal** (व्यवहार या राय जो सामान्य हैं).
 - **Vulnerable** means **capable to being attacked** (जो आघात योग्य हो)
 - **Peculiar** means **strange or unusual, sometimes in an unpleasant way** (अजीब या असामान्य, कभी-कभी एक अप्रिय तरीके से)
 - **Deify** means **considered to be a god or are regarded with very great respect** (एक देवता माना जाता है या बहुत सम्मान के साथ माना जाता है)

*Thus, the synonym of the word '**strange**' as used in the passage is option (3) **Peculiar**.*

Question 20

Read the following information carefully and answer the given questions.

Octopuses are famous for their alien-like abilities, from regrowing damaged arms to changing their skin colour and texture. They use this colour-shifting power to camouflage and, interestingly, as a strange visual language to talk to other octopuses.

A little known fact is they actually belong to a category of animals (phylum) called Mollusca, which is largely made up of snails. Yep, octopuses are like souped-up snails who lost their shells and grew a rather large brain. The greatest thing about them is their intelligence, which evolved completely independently from our own.

Let's consider their nervous system. Like us, octopus have large brains compared to their body size - easily the biggest of all invertebrates (animals without a backbone) and of comparable size to many vertebrates, such as frogs.

It is, however, hard to compare brain size between marine animals and land animals, because the laws of physics differ in water and air. Animals are weightless in water but on land body shape and size is limited by gravity. An octopus brain is made up of about 500 million brain cells (neurons). This is seven times more than a mouse and about the same as a marmoset monkey. Humans, on the other hand, have 86 billion brain cells.

Testing octopus intelligence can be a problem, because the animals frequently outsmart scientists. For example, scientists can struggle to get an octopus to solve a maze, because they often climb out and crawl over the top to reach their food reward. And that's assuming they haven't already escaped from their aquarium home and are crawling around the lab.

Unlike us though, octopuses don't live for very long. The giant Pacific octopus might live up to five years, but most live for just a year and some as little as six months. They hatch from eggs fully formed and ready to go. They never see their parents and have to learn everything on their own.

What can be an appropriate conclusion for the given passage?

Options:

- A. Octopuses have big brains and are crazy-smart.
- B. Octopuses change their behavior depending on the environment.
- C. Octopuses are alike humans in many ways.
- D. Octopuses don't live for very long.

Answer: A

Solution:

The correct answer is option (1) **Octopus have big brains and are crazy smart.**

Key Points

- Refer to the following lines of the passage: -
 - Line 2 of paragraph 3, "**Octopuses** are like souped-up snails who lost their shells and grew a rather **large brain**."
 - Line 3 of Para 3, "The greatest thing about them is their **intelligence**, which evolved completely independently from our own."
 - Line 1 of Para 5, "Testing **octopus intelligence** can be a problem, because the animals frequently **outsmart scientists**."
 - Thus, it can be derived that **option 1 is true and appropriate** as it **refers to the gist** of the passage.
 - Line 1 of Para 1, "Octopuses are famous for their alien-like abilities, from regrowing damaged arms to **changing their skin colour and texture**."
 - Thus, as per passage, Octopus **does not change behavior**. So, **option 2 is a false statement** as per passage.
 - Line 1 of Para 1, "Octopuses are famous for their **alien-like abilities**, from **regrowing damaged arms** to **changing their skin colour** and texture."
 - Line 1 of Para 2, "A little known fact is they actually belong to a **category of animals (phylum) called Mollusca**."
 - Line 2-4 of paragraph 4, "An **octopus brain** is made up of about **500 million** brain cells (neurons). This is seven times more than a mouse and about the same as a marmoset monkey. **Humans**, on the other hand, have **86 billion** brain cells."
 - Thus, **option 3 is a false statement** as per passage and Octopus are **not like humans**.
 - Line 1 of paragraph 6, "Unlike us though, octopuses **don't live for very long**."
 - Hence, **option 4 is also true but not an appropriate** conclusion because it refers to only one paragraph of the passage **not the gist** as per the given passage.

*Thus, the appropriate conclusion for the given passage will be **option (1) Octopus have big brains and are crazy smart**.*

RAGA

Question 21

Select the number that will come next in the following series.

2, 3, 6, 7, 14, 15, ?

Options:

A. 28

B. 32

C. 36

D. 30

Answer: D

Solution:

The pattern followed here is:



Hence, the next term in the series is 30.

Question 22

Arrange the following words in a logical and meaningful order.

1. Pupa
2. Death
3. Larva
4. Butterfly
5. Egg

Options:

- A. 3, 1, 2, 4, 5
- B. 3, 1, 4, 2, 5
- C. 5, 3, 1, 2, 4
- D. 5, 3, 1, 4, 2

Answer: D

Solution:

The logical order is as follows:

5) Egg

3) Larva

1) Pupa

4) Butterfly

2) Death

Hence, the correct answer is 5, 3, 1, 4, 2.

Question 23

Town S is 30 km to the North of Town T. Town U is 28 km to the East of Town T. Town V is 22 km to the West of Town S. If Sharman starts from Town V and goes to Town U, in which direction is he from the starting point?

Options:

A. South-East

B. South-West

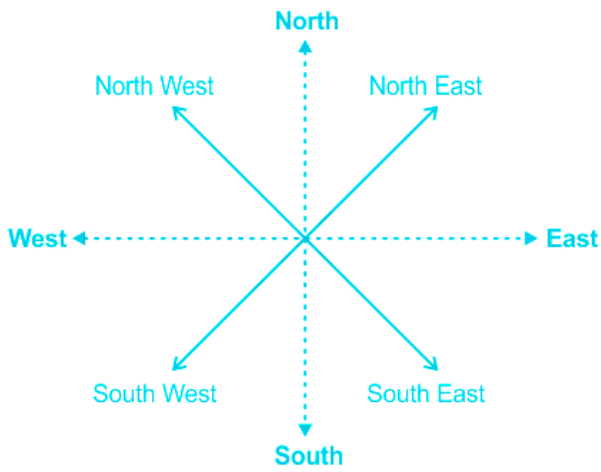
C. North-West

D. North-East

Answer: A

Solution:

The direction chart is as follows :-



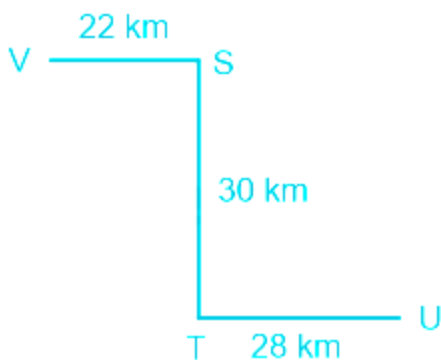
Step1:-Town S is 30 km to the North of Town T.



Step2:-Town U is 28 km to the East of Town T.



Step3:-Town V is 22 km to the West of Town S.



If Sharman start from Town V then he will go in south-east direction to reach Town U.

Hence, **the correct answer is south-east.**

Question 24

Six girls Mansi, Shweta, Monika, Payal, Aarti and Kajal are sitting around a circular table facing towards the centre (not necessarily in the same order). Kajal is to immediate left of Mansi. Shweta is third to the left of Mansi. Payal is second to the right of Kajal. Monika is not second to the left of Mansi. Who is fourth to the right of Aarti?

Options:

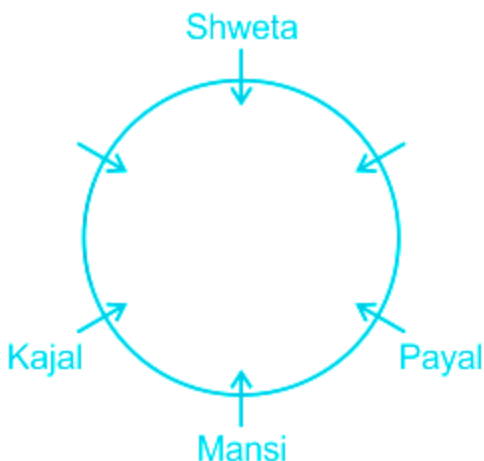
- A. Shweta
- B. Kajal
- C. Payal
- D. Monika

Answer: D

Solution:

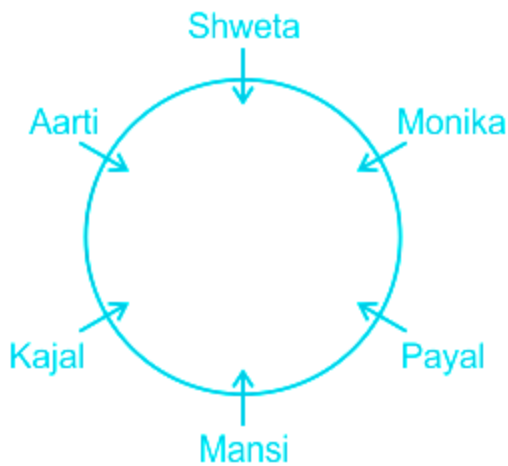
Given: Six girls Mansi, Shweta, Monika, Payal, Aarti and Kajal.

- 1) Kajal is to immediate left of Mansi.
- 2) Shweta is third to the left of Mansi.
- 3) Payal is second to the right of Kajal.



- 4) Monika is not second to the left of Mansi.

And, Aarti will sit at the remaining place.



Here, 'Monika' is fourth to the right of Aarti.

Hence, the correct answer is "**Option 4**".

Question 25

If the first half of the English alphabet series is reversed, which of the following will be the twelfth to the right of the seventh letter from the left end?

Options:

- A. P
- B. O
- C. S
- D. N

Answer: C

Solution:

We solve the given statement with the help of the following table:

Alphabets	A	B	C	D	E	F	G	H	I	J	K	L	M
Positional value	1	2	3	4	5	6	7	8	9	10	11	12	13
Positional value	26	25	24	23	22	21	20	19	18	17	16	15	14
Alphabets	Z	Y	X	W	V	U	T	S	R	Q	P	O	N

Step 1: We reverse the first half of the alphabets series,

MLKJIHG FEDCBANOPQRSTUVWXYZ

Step 2: Seventh letter from the left end.

MLKJIHGFEDCBANOPQRSTUVWXYZ

G is the seventh letter from the left end.

Step 3: The twelfth to the right of the seventh letter from the left end.

MLKJIHG FEDCBANOPQRSTUVWXYZ

S is twelfth to the right of the seventh letter from the left end

Hence the correct answer is option (3).

Question 26

In a certain code language ATTACK is written as ZGRCXP, then what is the code for BRANCH in same code language?

Options:

A.

SXYYI

B. YYISX

C. YYIXS

D. YIYPXS

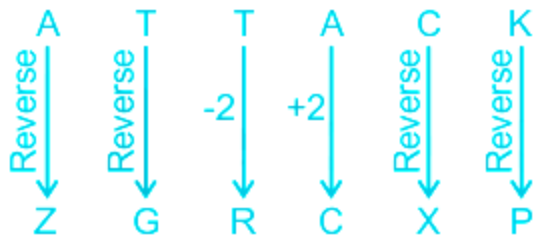
Answer: D

Solution:

According to the English alphabet series and its positional value:

Alphabets	A	B	C	D	E	F	G	H	I	J	K	L	M
Positional value	1	2	3	4	5	6	7	8	9	10	11	12	13
Positional value	26	25	24	23	22	21	20	19	18	17	16	15	14
Alphabets	Z	Y	X	W	V	U	T	S	R	Q	P	O	N

The pattern followed here is:



Similarly;



Hence, "YIYPXS" is the correct answer.

Shortcut Trick

Shortcut to remember the reverse positional letters.

A ↔ Z	(Azad)
B ↔ Y	(BoY)
C ↔ X	(CraX)
D ↔ W	(DeW)
E ↔ V	(EVEN)
F ↔ U	(FULL)
G ↔ T	(G T road)
H ↔ S	(High School)
I ↔ R	(Indian Railway)
J ↔ Q	(Jungle Queen)
K ↔ P	(KanPur)
L ↔ O	(LiOn)
M ↔ N	(MaN)

Question 27

At 8:30, the hour hand and the minute hand of a clock form an angle of

Options:

A. 80°

B. 75°

C. 70°

D. 60°

Answer: B

Solution:

Given :

Hour = 8

Minutes = 30

Formula used :

Angle between hour and minute hand:

$$\theta = \text{Hour} \times 30 - \frac{11}{2} \times \text{Minutes}$$

Calculations :

Using the above concept, we get

$$\Rightarrow \theta = 8 \times 30 - \frac{11}{2} \times 30$$

$$\Rightarrow \theta = 240 - (11 \times 15)$$

$$\Rightarrow \theta = 240 - 165$$

$$\Rightarrow \theta = 75^\circ$$

$\therefore$ At 8:30, the hour hand and the minute hand of a clock form an angle of 75° .

Question 28

Amongst six friends Uday, Vikrant, William, Xena, Yasir and Ziya, each has a different height. Vikrant is taller than only three other friends. Ziya is only taller than Xena and William is only shorter than Uday. Given that no two friends have the same height, find who is the third shortest among all the friends?

Options:

- A. Uday
- B. Ziya
- C. Yasir
- D. William

Answer: C

Solution:

Given: Six friends Uday, Vikrant, William, Xena, Yasir and Ziya.

Given that no two friends have the same height

1) Vikrant is taller than only three other friends.

___ > ___ > Vikrant > ___ > ___ > ___

2) Ziya is only taller than Xena and William is only shorter than Uday.

Uday > William > Vikrant > ___ > Ziya > Xena

So, Yasir will sit on the remaining place because, no two friends have the same height.

Uday > William > Vikrant > Yasir > Ziya > Xena

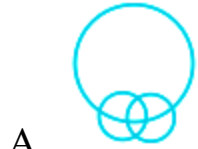
Here, 'Yasir' is the third shortest among all the friends.

Hence, the correct answer is "**Yasir**".

Question 29

Which diagram depicts relationship between Nitrogen, Ice, Air?

Options:

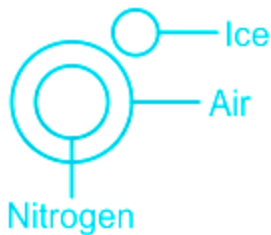


Answer: D

Solution:

Nitrogen is a part of air while ice is different from the two.

So the possible Venn diagram is:



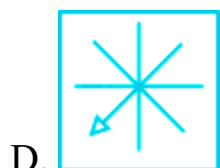
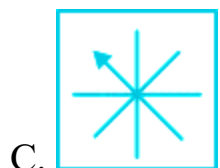
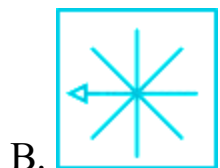
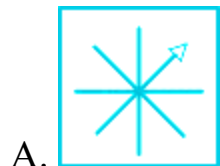
Hence, the figure in option 4 is the correct answer.

Question 30

Select the figure from the given options that can replace the question mark (?) in the following series.



Options:

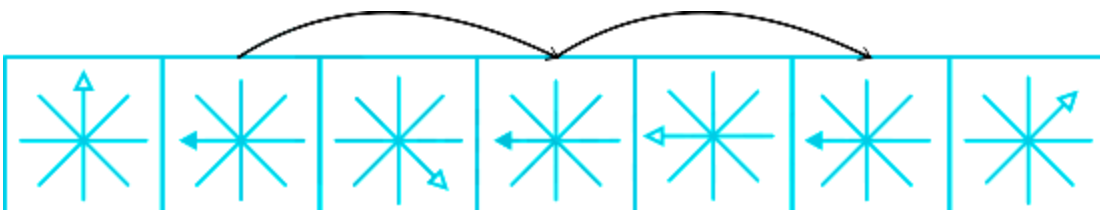


Answer: B

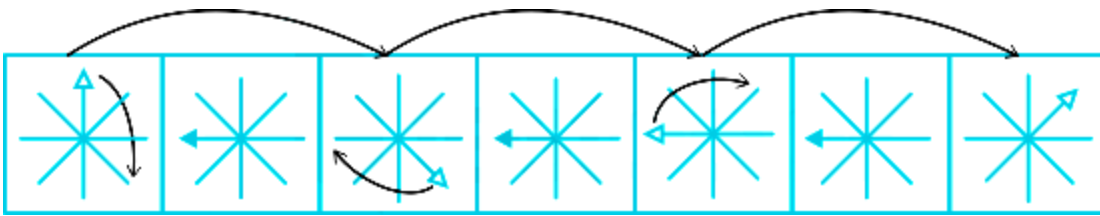
Solution:

The logic followed here is:

Step I: The shaded triangle is placed as it is in each alternate images.



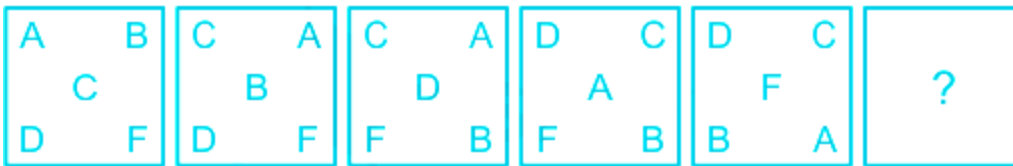
Step II: The empty triangle is moved three places in clockwise direction in each alternate images.



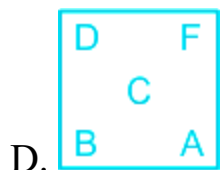
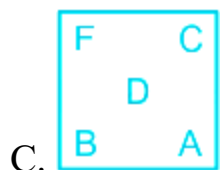
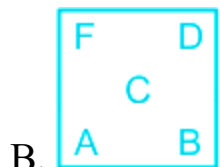
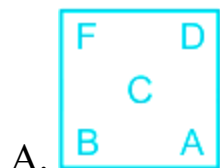
Hence, the correct answer is "Option 2".

Question 31

select the figure from the given options that can replace the question mark (?) in the following series.



Options:

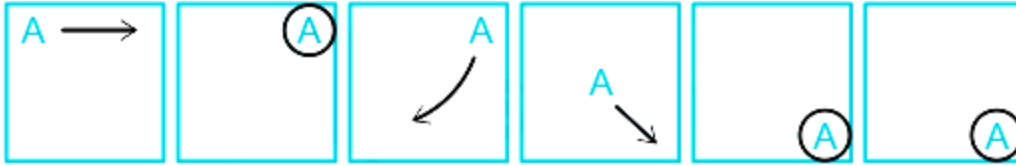


Answer: A

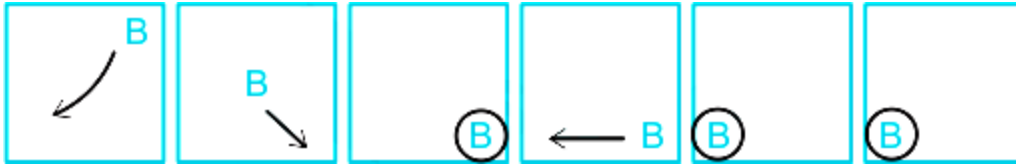
Solution:

The logic followed here is:

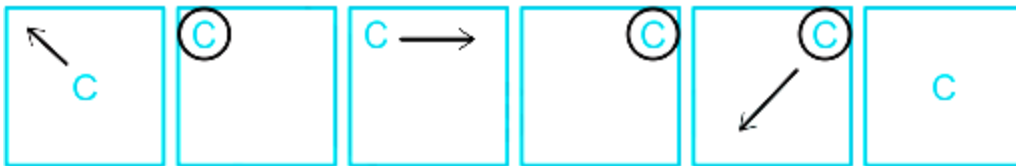
Step I: Letter A has shifted his place and keeping constant his place in four boxes.



Step II: Letter B has shifted his place and keeping constant his place in four boxes.



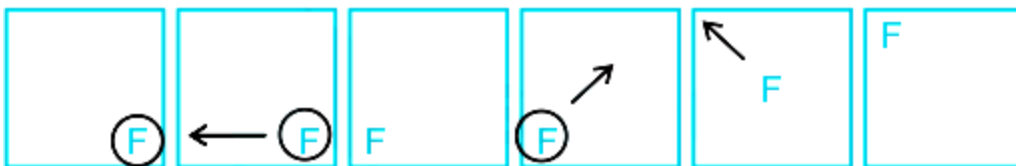
Step III: Letter C has shifted his place and keeping constant his place in four boxes.



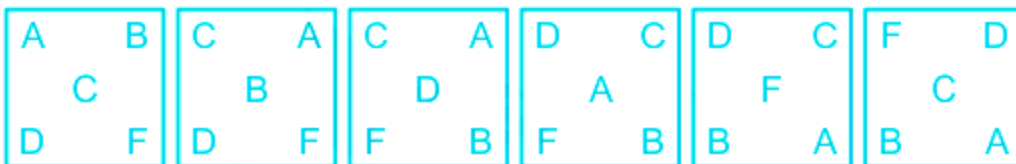
Step IV: Letter D has shifted his place and keeping constant his place in four boxes.



Step V: Letter F has shifted his place and keeping constant his place in four boxes.



So, the complete series is:



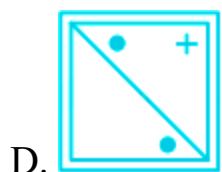
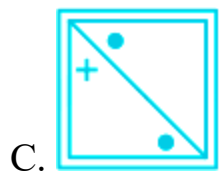
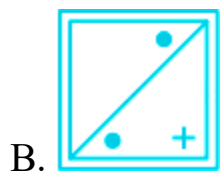
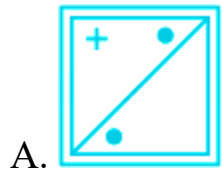
Hence, the correct answer is "Option 1".

Question 32



The next figure is

Options:



Answer: A

Solution:

Logic:

1). "+" is moving to the corners in anti clock wise direction.

2). Shaded dot is changing in to blank dot alternatively and moving from inside to outside on the diagonal line, which is also moving alternatively.

So, the next image will be;

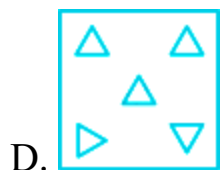
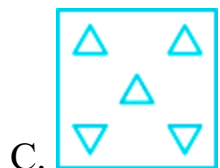
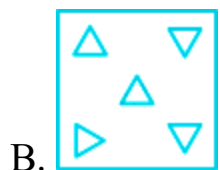
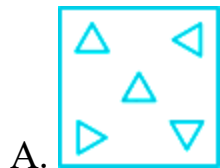


Question 33

Select the figure from the given options that can replace the question mark (?) in the following series.



Options:

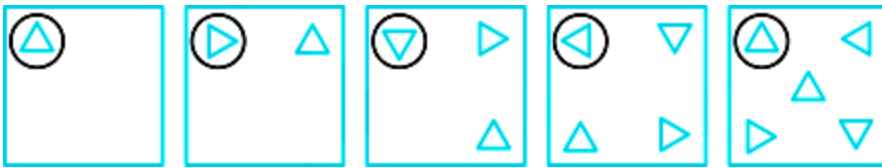


Answer: A

Solution:

The logic followed here is:

Logic: The Triangles are rotated by 90 degree in clockwise direction, and increase by an triangle in each next images.



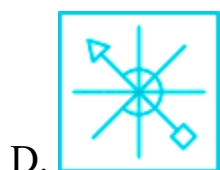
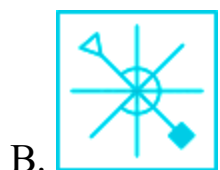
Hence, the correct answer is "**Option 1**".

Question 34

Select the figure from the given options that can replace the question mark (?) in the following series.



Options:



Answer: C

Solution:

The logic followed here is:

Step I: Symbol 'Square' has shifting twoplace in clockwise direction in next images, and getting Shaded in alternate image.



Step II: Symbol 'Triangle' has shifting twoplace in clockwise direction in next images, and getting Shaded in alternate image.



Step III: Symbol 'Circle' has subtracting one part in each next images.



So, the complete series will be:



So, 'Option 3' will replace the question mark.

Hence, the correct answer is "**Option 3**".

! Mistake Points

Both Option 1 and 3 has the different image, please check the shaded square carefully.

Question 35

Inquestion select the related word from given alternatives.

Bangladesh : Dhaka : : Germany : ?

Options:

- A. Paris
- B. Berlin
- C. Baghdad
- D. Rome

Answer: B

Solution:

The logic follows here is :-

Dhaka is capital of Bangladesh

Similarly,

Berlin is capital of Germany.

Thus, the correct answer is 'Berlin'.

Question 36

If, $\frac{50}{x} = \frac{x}{\frac{25}{2}}$, then the value of x is ?

Options:

- A. $\frac{25}{2}$
- B. $\frac{4}{25}$
- C. 4
- D. 25

Answer: D

Solution:

Calculation:

$$\frac{50}{x} = \frac{x}{\frac{25}{2}}$$

$$\Rightarrow \frac{50}{x} = \frac{x \times 2}{25}$$

$$\Rightarrow 50 \times 25 = 2x^2$$

$$\Rightarrow 25 \times 25 = x^2$$

$$\Rightarrow x = 25$$

$\therefore$ The correct answer is 25.

Question 37

Six years ago, the ratio of the ages of Kunal and Sagar was 6 : 5. Four years hence, the ratio of their ages will be 11 : 10. What is Sagar's age at present?

Options:

A. 16 years

B. 18 years

C. 20 years

D. can't determine

Answer: A

Solution:

Given

Six years ago, the ratio of the ages of Kunal and Sagar was 6 : 5. Four years hence, the ratio of their ages will be 11 : 10.

Concept used

Ratio and proportion

Calculation

Let the age of Kunal and Sagar 6 years ago be $6x$ and $5x$

Their present age will be $(6x + 6)$ and $(5x + 6)$

According to the question

Age of Kunal 4 year hence($6x + 10$)

Age of Sagar 4 year hence($5x + 10$)

Now, According to the question

$$\Rightarrow \frac{6x + 10}{5x + 10} = \frac{11}{10}$$

$$\Rightarrow 5x = 10$$

$$\Rightarrow x = 2$$

Sagar's present age

$$\Rightarrow 5 \times 2 + 6$$

$\therefore$ 16years.

Question 38

A student got 15% marks and failed by 95 marks. If he scores 35% marks then he gets 5 marks more than the passing marks. Find the passing marks.

Options:

A. 175

B. 150

C. 170

D. 160

Answer: C

Solution:

Given:

A student gets 15% marks and failed by 95marks.

Another student gets 35% marks and thus he scored 5marks more than the passing marks.

Concept used:

Using the concept of percentage

Calculation:

Let the total marks be 'x'.

According to the question,

$$15\% \text{ of } x + 95 = 35\% \text{ of } x - 5$$

$$\Rightarrow 35\% \text{ of } x - 15\% \text{ of } x = 95 + 5$$

$$\Rightarrow 20\% \text{ of } x = 100$$

$$\Rightarrow x = (100 \times 100) / 20$$

$$\Rightarrow x = 500$$

So, Passing marks = $35\% \text{ of } x - 5$

$$\Rightarrow (35 \times 500) / 100 - 5$$

$$\Rightarrow 175 - 5$$

$$\Rightarrow 170$$

$\therefore$ The passing marks is 170.

Question 39

A person takes $3\frac{1}{2}$ hours to cover a distance of 140 km. If he decreased his speed by 5 km/h, then what is the increase in time taken by him (in minutes) to cover the same distance ?

Options:

A. 18

B. 20

C. 30

D. 45

Answer: C

Solution:

Given:

Distance: 140 km Initial time taken: $7/2$ hours

Decrease in speed: 5 km/h

Concept used:

The formula relating speed, distance, and time is: $\text{Speed} = \text{Distance} / \text{Time}$

Calculation:

Initial speed = Distance / Time = $140 \text{ km} / (7/2) \text{ hours} = 40 \text{ km/h}$

Decreased speed = Initial speed - 5 km/h = $40 \text{ km/h} - 5 \text{ km/h} = 35 \text{ km/h}$

Time taken with decreased speed = Distance / Decreased speed = $140 \text{ km} / 35 \text{ km/h} = 4 \text{ hours}$

Increase in time = 4 hours - $(7/2) \text{ hours}$

$\Rightarrow$ Increase in time = $1/2 \text{ hours}$

$\Rightarrow$ Increase in time = 30 minutes

$\therefore$ The increase in time taken by him to cover the same distance is 30 minutes.

**Alternate Method****Calculation:**

Initial speed = Distance / Time = $140 \text{ km} / (7/2) \text{ hours} = (140 \times 2) / 7 = 40 \text{ km/h}$

New speed = Initial speed - Decrease in speed = $40 \text{ km/h} - 5 \text{ km/h} = 35 \text{ km/h}$

To find the time taken at the new speed, we divide the distance by the new speed:

$\Rightarrow$ New time taken = Distance / New speed = $140 \text{ km} / 35 \text{ km/h} = 4 \text{ hours}$

The increase in time taken is the difference between the initial time and the new time:

$\Rightarrow$ Increase in time taken = New time taken - Initial time taken = $4 \text{ hours} - (7/2) \text{ hours}$

To convert the time difference to minutes, we multiply by 60:

$\Rightarrow$ Increase in time taken (in minutes) = $(4 - (7/2)) \text{ hours} \times 60 \text{ minutes/hour}$

$\Rightarrow$ Calculating the expression: Increase in time taken (in minutes)

$= (8 - 7) / 2 \times 60 = 1/2 \times 60 = 30 \text{ minutes}$

Therefore, the increase in time taken by the person to cover the same distance is 30 minutes.

Question 40

The L.C.M. of $\frac{2}{4}$, $\frac{5}{6}$, $\frac{10}{8}$ is:

Options:

A. $\frac{1}{5}$

B. $\frac{5}{4}$

C. $\frac{4}{5}$

D. $\frac{5}{2}$

Answer: D

Solution:

Concept used:

LCM of Fraction = LCM of Numerator/HCF of Denominator

Calculation:

$$\frac{2}{4}, \frac{5}{6}, \frac{10}{8} = \frac{1}{2}, \frac{5}{6}, \frac{5}{4}$$

$$\Rightarrow \text{LCM of } (1, 5, 5) = 5$$

$$\Rightarrow \text{HCF of } (2, 6, 4) = 2$$

$$\Rightarrow \frac{\text{LCM of } (1, 5, 5)}{\text{HCF of } (2, 4, 6)} = \frac{5}{2}$$

$\therefore$ The correct answer is $5/2$.

Mistake Points

Please note that LCM means the lowest common multiple. LCM is the lowest number which is completely divisible by all given numbers ($2/4$, $5/6$, $10/8$).

In these types of questions, make sure that you reduce the fractions to their lowest forms before you use their formulae, otherwise, you may get the wrong answer.

If we don't reduce the fractions to their lowest forms then LCM is 5 but the LCM of these 3 numbers is $5/2$.

Question 41

The average of the first 43 natural numbers is:

Options:

A. 33

B. 44

C. 43

D. 22

Answer: D

Solution:

Given:

First 43 natural numbers

Concept Used:

Average of n natural numbers = $\frac{n+1}{2}$

Calculation:

$$\Rightarrow \text{Average of first 43 natural numbers} = \frac{43+1}{2} = \frac{44}{2}$$

$$\Rightarrow \text{Average} = 22$$

Hence, the average of first 43 natural numbers is 22

Question 42

If $\sin \theta = \frac{5}{13}$, then what is the value of $\frac{\cos^2 \theta - \sin^2 \theta}{2 \cos \theta \cdot \sin \theta}$?

Options:

A. 115/126

B. 113/120

C. 117/136

D. 119/120

Answer: D

Solution:

Given:

$$\sin \theta = \frac{5}{13}$$

Concept Used:

$$\cos \theta = \sqrt{1 - \sin^2 \theta}$$

Calculation:

$$\sin \theta = 5/13$$

$$\Rightarrow \cos \theta = \sqrt{1 - (5/13)^2}$$

$$\Rightarrow \cos \theta = \sqrt{(169 - 25)/169}$$

$$\Rightarrow \cos \theta = \sqrt{144/169}$$

$$\Rightarrow \cos \theta = 12/13$$

Put the values in $\frac{\cos^2 \theta - \sin^2 \theta}{2 \cos \theta \cdot \sin \theta}$

$$\Rightarrow \frac{(12/13)^2 - (5/13)^2}{2 \times 12/13 \times 5/13}$$

$$\Rightarrow \frac{119/169}{120/169}$$

$$\Rightarrow 119/120$$

∴ Option 4 is the correct answer.

Question 43

Which of the following player has received the Major Dhyan Chand Khel Ratna Award 2022?

Options:

A. Manika Batra

B. Sathiyar Gnanasekaran

C. Harmeet Desai

D. Sharath Kamal Achanta

Answer: D

Solution:

The correct answer is Sharath Kamal Achanta.

Key Points

- **Achanta Sharath Kamal** to be honored with Major Dhyan Chand Khel Ratna Award 2022.
- Achanta Sharath Kamal is an Indian professional **table tennis player**.
- He is the first Indian table tennis player ever to become **ten times Senior National Champion** hence breaking the record of **eight times National Champion Kamlesh Mehta**.
- In **2019** he was awarded the **Padma Shri**, India's fourth-highest civilian award.

Additional Information

- **Top five Indian players at World Team Table Tennis Championships:**
 - Sathiyam Gnanasekaran
 - Sreeja Akula
 - Harmeet Desai
 - Manika Batra
 - Manav Thakkar
 - Table tennis, also known as **ping-pong and whiff-whaff**, is a sport in which two or four players hit a lightweight ball.
 - **Fan Zhendong** is a **Chinese professional table tennis player** who is currently ranked **world No. 1** for men's singles by the International Table Tennis Federation.
-

Question 44

When was the MGNREGA Scheme passed?

Options:

A. 2005

B. 2008

C. 2006

D. 2007

Answer: A

Solution:

The correct answer is 2005.



Key Points

- The **MGNREGA** was passed on **23rd August 2005**.
- MGNREGA stands for **Mahatma Gandhi National Rural Employment Guarantee Act**.
- The act was first proposed in **1991** by **P.V. Narasimha Rao**.
- It was **finally accepted in the parliament** and commenced implementation in **625 districts of India**.



Additional Information

- On **2nd October 2009**, an amendment was made in the **National Rural Employment Guarantee Act 2005**, to **change the nomenclature of the Act from NREGA to MGNREGA**.
 - The primary objective of **MGNREGA** is to **guarantee 100 days of wage employment** per year to rural households.
-

Question 45

Pradhan Mantri Fasal Bima Yojna was launched in the year:

Options:

- A. 2013
- B. 2014
- C. 2016
- D. 2017

Answer: C

Solution:

The correct answer is option 3, i.e. 2016.

- Pradhan Mantri Fasal Bima Yojna was launched in the year 2016.
- The Pradhan Mantri Fasal Bima Yojana (PMFBY) was launched on **18 February 2016**.
- It is an insurance service for farmers for their yields.
- It was formulated in line with **One Nation–One Scheme** theme.

- It replaces the two schemes National Agricultural Insurance Scheme (NAIS) and the Modified National Agricultural Insurance Scheme.
 - It aims to reduce the premium burden on farmers.
 - It also ensures the early settlement of crop assurance claim for the full insured sum.
-

Question 46

Why did Gandhiji start the Champaran Satyagraha?

Options:

- A. To increase the wages of the textile mill workers
- B. To reduce the taxes given by the farmers
- C. To protest against British exploitation of the Indigo farmers
- D. None of the above

Answer: C

Solution:

Mohandas Karamchand Gandhi (2 October 1869– 30 January 1948) was an **Indian lawyer, anti-colonial nationalist, and political ethicist**, who employed nonviolent resistance to lead the successful campaign for India's independence from British rule, and in turn, inspired movements for civil rights and freedom across the world.



Key Points

Champaran Satyagraha:

- The Champaran Satyagraha of **1917** was the **first Satyagraha movement** led by Gandhiji in India and is considered a historically important revolt in the Indian Independence Movement.
- It was a farmer's uprising that took place in the Champaran district of Bihar, India, during the British colonial period. The **farmers were protesting against having to grow indigo with barely any payment for it.**
- Champaran, where tens of thousands of landless serfs, indentured labourers, and poor farmers were forced to grow indigo and other cash crops instead of the food crops which were necessary for their survival.
- These goods were bought from the farmers at a very low price. Suppressed by the brutal militias of the landlords, they were given negligible compensation, leaving them in extreme poverty.
- Even in the painful time of devastating famine, the **British government levied a heavy tax on them and insisted on increasing the rate.**
- Without food and money, the situation was growing progressively unbearable and the peasants in Champaran revolted against the government in indigo plant cultivation in 1914 (at Pipra) and in 1916 at (Turkaulia).
- Raj Kumar Shukla who was an indigo cultivator persuaded Mahatma Gandhi to go to Champaran and hence, the Champaran Satyagraha began. Gandhiji arrived in Champaran on 10 April 1917 with a team of eminent lawyers i.e. Brajkishore Prasad, Rajendra Prasad, Anugrah Narayan Sinha, and Acharya Kripalani.

Hence, the correct answer is to protest against British exploitation of the Indigo farmers.

Question 47

Why does a black board appears black in colour?

Options:

- A. It reflects black colour
- B. It absorbs black colour
- C. It reflects all colours
- D. It absorbs all the colours

Answer: D

Solution:

The board absorbs all light that falls on it so it appears as "Black", which is actually the absence of colour. However, no substance is a perfect black body so it will reflect some of the light incident on it whose intensity is so low that it cannot be noticed by the naked eye.

Question 48

The lowest layer of atmosphere is called

Options:

- A.
Thermosphere
- B.
Troposphere
- C.
Mesosphere

D.

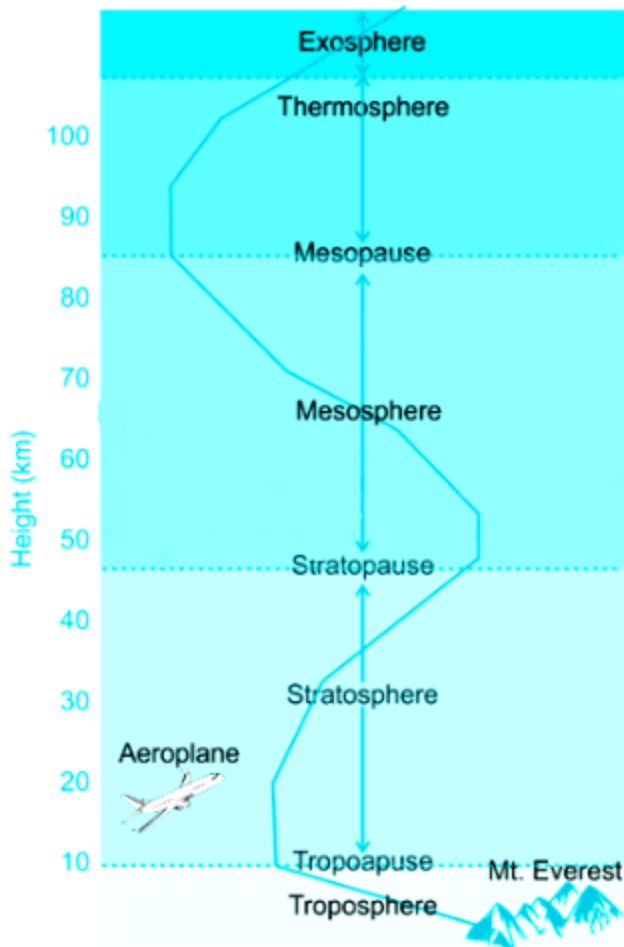
Ionosphere

Answer: B

Solution:

Explanation:

- The correct answer is **Troposphere**
- The troposphere is the **lowest layer** of the Earth's atmosphere.
- The atmosphere is comprised of layers based on temperature.
- These layers are the troposphere, stratosphere, mesosphere, and thermosphere.
- A further region at about 500 km above the Earth's surface is called the exosphere.



Question 49

What is the process of transferring files from internet to computer memory called?

Options:

- A. Uploading
- B. Downloading
- C. Bookmarking
- D. Hyperlinking

Answer: B

Solution:

- **Downloading** is the process of transferring files from internet to computer memory.
 - In other words, it means to receive data from a remote system, typically a server.
-

Question 50

The length breadth and height of a cuboid are in the ratio 4: 3: 2. If the volume of the cuboid is 1536 cm^3 , then what will be the total surface area of the cuboid?

Options:

- A. 868 cm^2
- B. 416 cm^2
- C. 624 cm^2
- D. 832 cm^2

Answer: D

Solution:**Formula used:**

The volume of cuboid = $L \times B \times H$

Total surface area of cuboid = $2(LB + BH + HL)$

Calculation:

$$L \times B \times H = 1536$$

Let the length breadth and height of a cuboid are in the ratio $4x: 3x: 2x$

$$\therefore 24x^3 = 1536$$

$$1536/24 = 64 = x^3 \text{ or } x = 4$$

$$\text{Total surface area of cuboid} = 2 \times (12x^2 + 6x^2 + 8x^2) = 52x^2$$

$$\therefore \text{Total surface area of cuboid} = 52 \times 16 = 832 \text{ cm}^2$$
